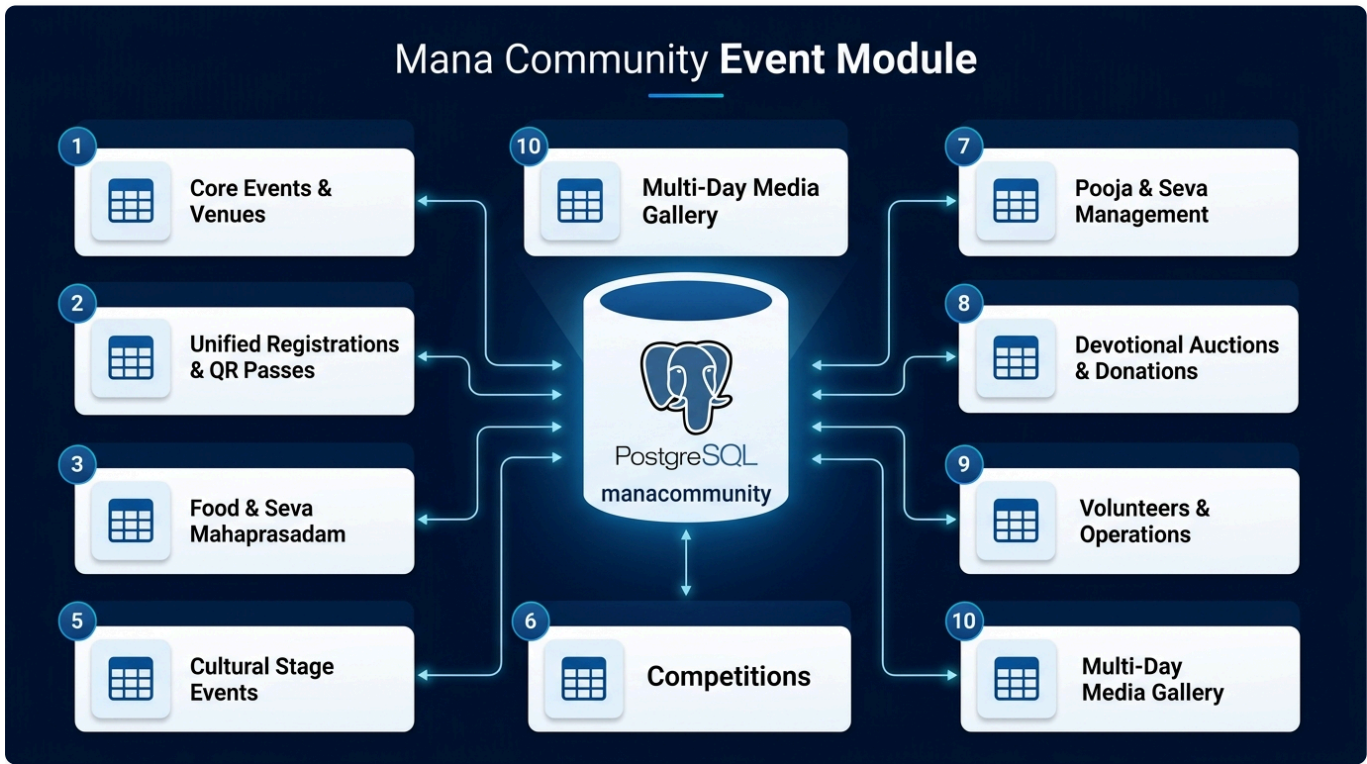
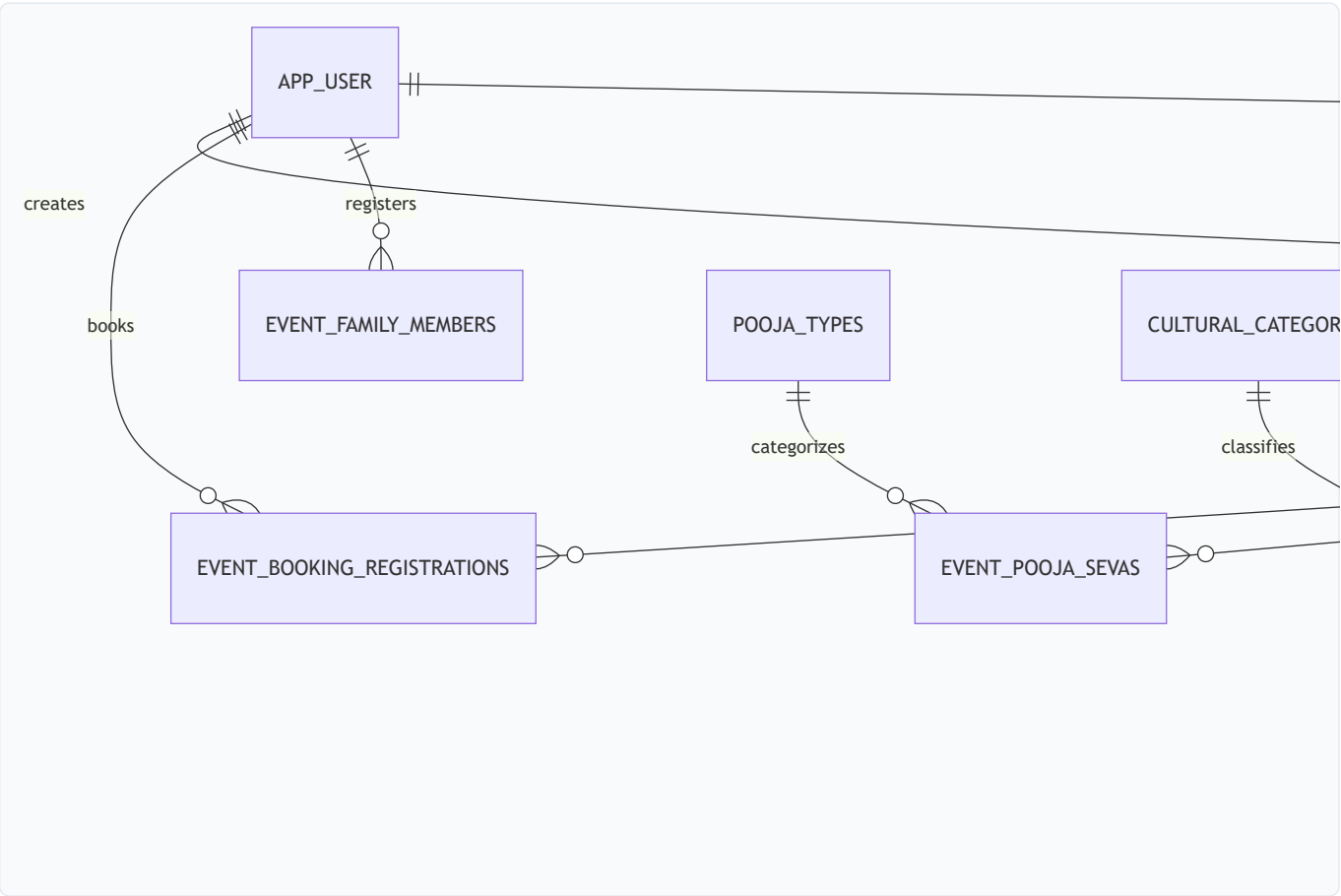


Mana Community App — Event Module Database Architecture & Schema Blueprint

Database System: PostgreSQL 15+ / RDS
 Schema Name: manacommunity
 ORM Layer: Spring Data JPA / Hibernate 6.x
 Migrations: Flyway (v1__baseline through v58__add_payment_and_contact_details)
 Backend Service: mana-community-service
 Frontend App: mana-community-app (React 19 / Vite / TailwindCSS)



High-Level Architecture & Entity Relationship Diagram

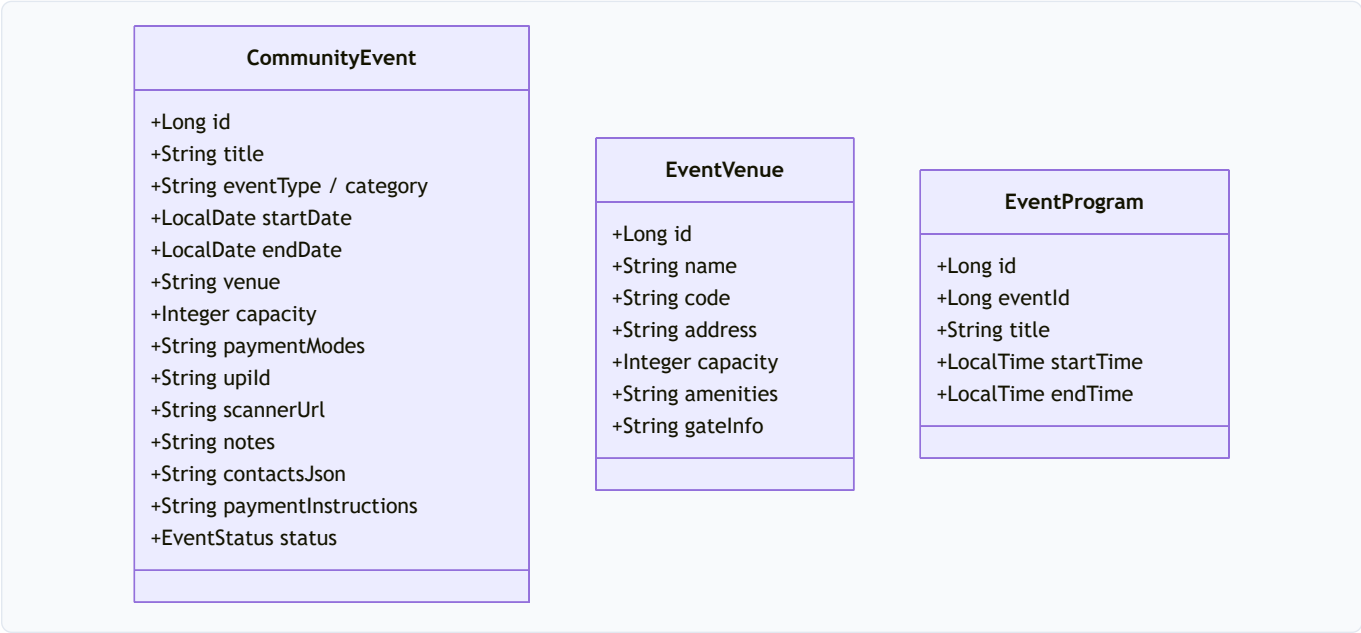


Summary of Event Subsystems (28 Tables)

#	FUNCTIONAL SUBSYSTEM	TABLE COUNT	PRIMARY RESPONSIBILITIES
1	Core Events & Venues	3	Master festival events, venue facilities, program schedules
2	Unified Registrations & Passes	4	Registrations, E-passes, QR codes, devotee family lists
3	Pooja & Seva Management	2	Seva rituals, daily time slots, pandit info, Gotram chanting
4	Food & Mahaprasadam	2	Catering sessions, plate counts, dining token bookings
5	Cultural & Stage Events	3	Performances, classical categories, Solo/Group formats
6	Competitions & Contests	3	Community contests, age tiers (Kids/Youth/Open), scoring
7	Auctions, Donations & Sponsors	4	Laddu/Idol auctions, live bidding, hundi donations, sponsors
8	Finance, Invoices & Expenses	2	Budget vs actual expenses, decorator & audio vendor invoices
9	Volunteers & Task Operations	2	Volunteer rosters, shift duties, organizer to-do action items
10	Media, Timeline & Gallery	3	Multi-day festival timeline, photo categories, gallery media

Complete Subsystem & Table Data Dictionary

Subsystem 1: Core Events, Agendas & Venues



1.1 community_event

- Entity: `CommunityEvent.java`
- Description: Central event records supporting multi-day festivals, ticketing, online/manual payment configuration, QR scanners, and committee contacts.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	BIGSERIAL	PRIMARY KEY	Unique event identifier
title	VARCHAR(255)	NOT NULL	Event / Festival title (e.g. <i>Ganesh Chaturthi Utsav 2026</i>)
description	TEXT	NULL	Full description and schedule details
type / category	VARCHAR(50)	DEFAULT 'festival'	Type: festival , pooja , cultural , sports , general
start_date / end_date	DATE	NOT NULL	Multi-day festival start & end dates
start_time / end_time	TIME	NULL	Daily operating hours
location / venue	VARCHAR(255)	NULL	Primary venue/hall name
venue_id	BIGINT	FK -> event_venues(id)	Linked structured venue
capacity / max_attendees	INT	NULL	Total capacity / registrant limit
price_type	VARCHAR(20)	DEFAULT 'FREE'	FREE , PAID , DONATION
price	NUMERIC(10,2)	NULL	Base pass fee (if single tier)
payment_modes	VARCHAR(255)	NULL	Comma-delimited: UPI , Card , NetBanking , Cash
upi_id	VARCHAR(100)	NULL	Event VPA (temple.trust@icici)
scanner_url	VARCHAR(1000)	NULL	S3 URL or direct link to payment QR code scanner image
notes	TEXT	NULL	Devotee guidelines, dress code, parking & entry instructions
contacts_json	TEXT	NULL	JSON array of committee coordinators [{name, phone, role}]
payment_instructions	TEXT	NULL	Instructions for UPI scan & UTR confirmation
status	VARCHAR(30)	DEFAULT 'PUBLISHED'	DRAFT , PUBLISHED , COMPLETED , CANCELLED
created_by	BIGINT	FK -> app_user(id)	Creator / Organizer user ID
community_id	BIGINT	FK -> community(id)	Tenant community ID

1.2 event_venues

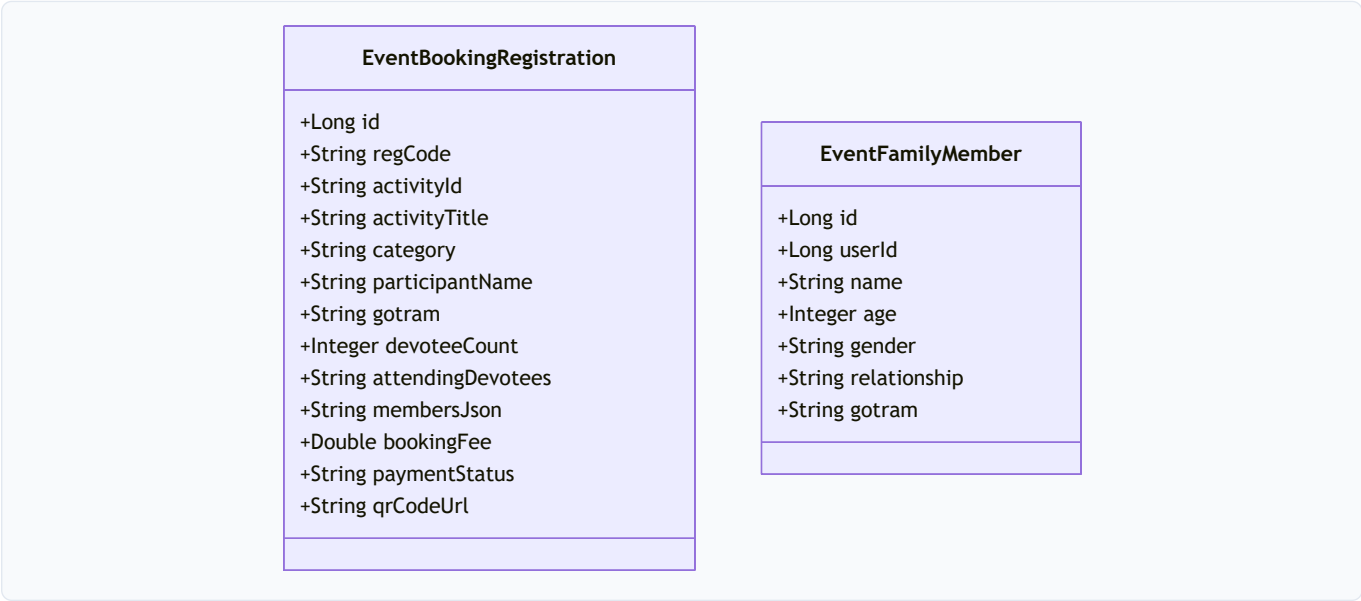
- Entity: `EventVenue.java`
- Description: Physical community venues, halls, grounds, gate details, and amenities.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	BIGSERIAL	PRIMARY KEY	Venue ID
name	VARCHAR(255)	NOT NULL	Venue name (e.g. <i>Main Temple Mandap</i>)
code	VARCHAR(50)	UNIQUE	Short code (MANDAP-01 , HALL-A)
address / city / state	VARCHAR(255)	NULL	Physical location info
capacity	INT	NULL	Seating / standing capacity
amenities	TEXT	NULL	AC, Audio System, Stage Lighting, Generator backup
gate_info	VARCHAR(255)	NULL	Entry gate directions (e.g. <i>Gate 2 - North Entrance</i>)

1.3 event_program

- Entity: EventProgram.java
- Description: Detailed timeline and agenda items scheduled across the event dates.

Subsystem 2: Unified Registrations, Passes & Family Attendees



2.1 event_booking_registrations

- Entity: EventBookingRegistration.java
- Description: Primary unified registration entity storing pass purchases, pooja bookings, devotee family member arrays, Gotram details, payment receipts, and QR check-in codes.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	BIGSERIAL	PRIMARY KEY	Registration record ID
reg_code	VARCHAR(100)	UNIQUE, NOT NULL	Human-readable code (P00JA-1001 , MNA-2026-EVENT-501)
activity_id	VARCHAR(100)	NULL	Sub-activity identifier (pooja-1 , food-2 , comp-3)
activity_title	VARCHAR(255)	NOT NULL	Title of registered activity or event
category	VARCHAR(50)	NULL	Category (Pooja , Cultural , Competition , Event)
user_id	BIGINT	FK -> app_user(id)	Registered user account
community_id	BIGINT	FK -> community(id)	Community tenant ID
participant_name	VARCHAR(255)	NOT NULL	Primary devotee / registrant full name
gotram	VARCHAR(100)	NULL	Gotram / Family lineage for ritual Sankalpam
devotee_count	INT	DEFAULT 1	Total attendees count (Self + Family members)
attending_devotees	TEXT	NULL	Comma-separated or JSON list of attendee names
members_json	TEXT	NULL	Full JSON array of attendees [{name, age, gender, rel}]
event_date / event_time	VARCHAR(100)	NULL	Scheduled ritual/event slot time
venue	VARCHAR(255)	NULL	Allocated venue or mandap
booking_fee	NUMERIC(10,2)	DEFAULT 0.0	Total contribution/fee paid
payment_status	VARCHAR(30)	DEFAULT 'PAID'	PAID , PENDING , FREE , REFUNDED
payment_method	VARCHAR(50)	NULL	UPI , Card , NetBanking , Cash Desk , Free Seva
payment_receipt_url	VARCHAR(1000)	NULL	S3 screenshot upload URL for payment proof
transaction_id	VARCHAR(100)	NULL	UPI UTR / Bank reference number
qr_code_url	VARCHAR(500)	NULL	Dynamic digital E-Pass QR code image URL
status	VARCHAR(30)	DEFAULT 'CONFIRMED'	CONFIRMED , PENDING APPROVAL , CANCELLED

2.2 event_family_members

- Entity: EventFamilyMember.java
- Description: Reusable family profiles linked to residents for 1-click inclusion in rituals and events.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	BIGSERIAL	PRIMARY KEY	Family member ID
user_id	BIGINT	FK -> app_user(id)	Primary resident account
name	VARCHAR(255)	NOT NULL	Member full name
age	INT	NOT NULL	Member age
gender	VARCHAR(20)	NOT NULL	Male , Female , Other
relationship	VARCHAR(50)	NOT NULL	Spouse , Child , Parent , Self (Head)
gotram	VARCHAR(100)	NULL	Associated gotram

2.3 event_registration & community_event_activity_registration

- **Entities:** [EventRegistration.java](#) , [ActivityRegistration.java](#)
- **Description:** Legacy and normalized activity registration join tables.

Subsystem 3: Pooja & Seva Management

3.1 event_pooja_sevas

- **Entity:** [PoojaSeva.java](#)
- **Description:** Sacred seva offerings, morning/evening ritual slots, pricing, and pandit assignments.

COLUMN	TYPE	DESCRIPTION
id	BIGSERIAL	Unique seva offering ID
main_event_id	BIGINT	FK -> community_event(id)
name	VARCHAR(255)	Seva Title (e.g. <i>Maha Ganapathi Abhishekam</i> , <i>Satyantarayan Puja</i>)
type	VARCHAR(100)	Seva type classification
date / end_date	VARCHAR(50)	Single day or multi-day ritual date range
is_multi_day	BOOLEAN	Supports multi-day homams
start_time / start_times	VARCHAR(255)	Slot timings (e.g. <i>08:30</i> , <i>11:00</i>)
duration	INT	Ritual duration in minutes
mandap	VARCHAR(255)	Dedicated Mandap/Homa Kund area
pandit	VARCHAR(255)	Head Priest / Pandit in charge
slots	INT	Remaining available booking capacity
fee	NUMERIC(10,2)	Seva booking fee (or <i>0</i> for free seva)
is_free	BOOLEAN	Free seva flag
items_json	TEXT	Required samagri list (Coconut, Flowers, Kumkum)

3.2 event_pooja_types

- **Entity:** [PoojaType.java](#)
- **Description:** Master taxonomy of Vedic pooja types (*Ganapati Homam* , *Abhishekam* , *Maha Aarti* , *Laghu Rudra* , *Sahasranama Archana*).

Subsystem 4: Food, Catering & Mahaprasadam

4.1 event_lunch_dinners

- **Entity:** [LunchDinner.java](#)
- **Description:** Community dining, Annadanam, and Mahaprasadam distribution sessions.

COLUMN	TYPE	DESCRIPTION
id	BIGSERIAL	Meal session record ID
main_event_id	BIGINT	FK -> community_event(id)
meal_type	VARCHAR(50)	Breakfast , Lunch , Dinner , Mahaprasadam
date / time	VARCHAR(50)	Serving date & time window
target_plates	INT	Planned plate preparation capacity
booked_plates	INT	Current count of confirmed devotee plates
venue	VARCHAR(255)	Dining Hall / Prasadam Counter
menu_json	TEXT	Traditional menu items served

4.2 community_event_meal_registration

- **Entity:** [MealRegistration.java](#)
- **Description:** Resident food tokens, plate quantities, and dietary preferences.

Subsystem 5: Cultural Events & Stage Performances

5.1 event_cultural_events

- **Entity:** [CulturalEvent.java](#)

- **Description:** Cultural stage activities, dance, music, drama, and youth performance programs.

COLUMN	TYPE	DESCRIPTION
id	BIGSERIAL	Performance event ID
main_event_id	BIGINT	FK -> community_event(id)
title	VARCHAR(255)	Performance title (e.g. <i>Bharatanatyam Pushpanjali</i>)
category	VARCHAR(100)	Classical Dance, Vocal, Bhajans, Skit
performance_type	VARCHAR(50)	Solo, Duet, Group (3-8), Mega Group (8+)
stage / venue	VARCHAR(255)	Main Stage, Temple Mandap, Auditorium
date / time	VARCHAR(50)	Performance schedule
duration	INT	Allocated stage time in minutes
coordinator	VARCHAR(255)	Cultural committee coordinator contact

5.2 event_cultural_categories & event_cultural_performance_types

- **Entities:** CulturalCategory.java , CulturalPerformanceType.java
- **Description:** Master categories (Classical Dance , Folk Dance , Carnatic Vocal , Instrumental , Drama) and format types.

Subsystem 6: Competitions & Contests

6.1 event_competitions

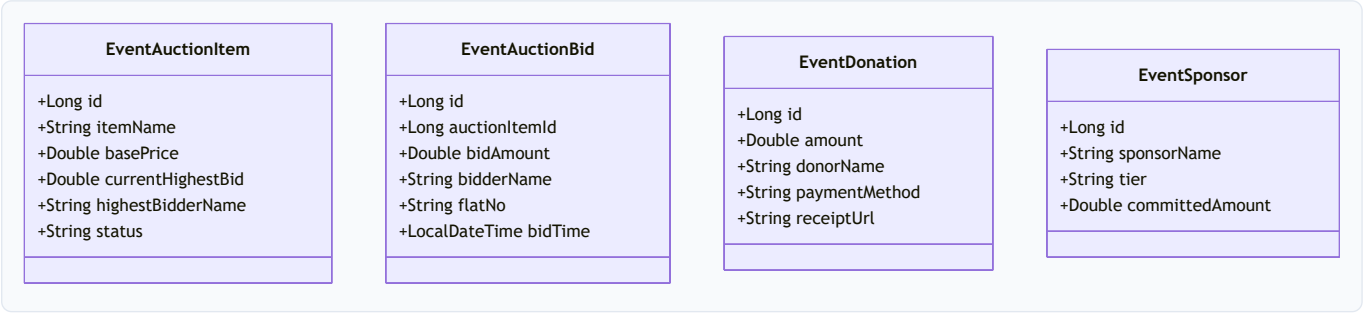
- **Entity:** Competition.java
- **Description:** Society talent contests, drawing, elocution, quiz, and sports tournaments.

COLUMN	TYPE	DESCRIPTION
id	BIGSERIAL	Competition record ID
main_event_id	BIGINT	FK -> community_event(id)
title	VARCHAR(255)	Contest title (e.g. <i>Drawing Competition – Kids</i>)
category	VARCHAR(100)	Art, Literary, Music, General Knowledge
age_group	VARCHAR(50)	Kids (5-10) , Junior (11-15) , Youth (16-25) , Open
fee	NUMERIC(10,2)	Entry registration fee
max_participants	INT	Participant cap / quota
prizes_json	TEXT	1st, 2nd, 3rd prize awards and certificates

6.2 event_competition_categories & event_competition_age_groups

- **Entities:** CompetitionCategory.java , CompetitionAgeGroup.java
- **Description:** Standardized age brackets and category taxonomies.

Subsystem 7: Auctions, Donations & Sponsors



7.1 event_auction_item & event_auction_bid

- **Entities:** EventAuctionItem.java , EventAuctionBid.java
- **Description:** Live devotional auction management (e.g. *Maha Laddu Prasadam Auction, Silver Idol Auction*), real-time bid logging, and winner declarations.

7.2 event_donation & community_event_sponsor

- **Entities:** EventDonation.java , EventSponsor.java
- **Description:** Voluntary hundi donations, 80G receipts, and corporate/vendor sponsorship contracts.

Subsystem 8: Finance, Invoices & Expenses

8.1 community_event_expense & event_invoice

- **Entities:** EventExpense.java , EventInvoice.java
- **Description:** Expense logging (Stage decoration, sound systems, pandit dakshina, flower vendors) and budget reconciliation against ticket/donation collections.

Subsystem 9: Volunteers & Task Operations

9.1 community_event_volunteer & community_event_task

- **Entities:** EventVolunteer.java , EventTask.java
- **Description:** Resident volunteer rosters (Security, Prasadam distribution, Stage coordination) and committee action items with priority tracking.

Subsystem 10: Gallery, Multi-Day Timeline & Media

10.1 event_media_day , event_media_category & event_gallery_item

- **Entities:** EventMediaDay.java , EventMediaCategory.java , EventGalleryItem.java
- **Description:** Photo and video galleries grouped by festival day (e.g. Day 1 - Sthapana, Day 5 - Cultural Night, Day 11 - Visarjan) and uploaded to S3.

 Security & Multi-Tenancy Design

1. **Community Isolation:** All tables contain community_id foreign keys indexed for sub-millisecond multi-tenant query filtering.
2. **User Authentication:** Binds user_id to app_user with @AuthenticationPrincipal validation in all controllers.
3. **Data Integrity:** Uses @PrePersist and @PreUpdate JPA entity hooks to manage audit timestamps (created_at , updated_at).
4. **Flyway Migration Strategy:** Schema changes are revision-controlled under src/main/resources/db/migration/ (V1 to V58).